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JAVA PROJECT REPORT ON

INCOME TAX ESTIMATION AND DEDUCTION CALCULATION SYSTEM

Submitted by

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DECLARATION

I hereby declare that the project entitled income tax estimation and deduction calculation **System"** submitted in partial fulfillment of the requirements for the course Java programming is a record of my original work carried out under the guidance of Mr.ARAVID, Technical Trainer, Technical Department, KG College of Arts and Science. I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution. I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

HINDHUPRIYA.M

CERTIFICATE

This is to certify that the project entitled "INCOME TAX ESTIMATION AND DEDUCTION CALCULATION SYSTEM" carried out by Hindhupriya.mIBSC.CSBBsc computer science. KG College of Arts & Science, in partial fulfillment of the requirements for the course java programming

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This project has been completed under my supervision and guidance during the academic

period **April 2026**.

Place: SARAVANAMPATTI

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ABSTRACT

The Income Tax Estimation and Deduction Calculation System is a Java-based application used to calculate income tax easily. It helps users by taking their income and deduction details as input and automatically calculating the taxable income and tax amount. This system reduces errors and makes tax calculation simple and fast. It is developed using basic Java concepts like conditions and loops, and it can be used as a console program or with a simple interface.

The system is developed using basic Java concepts like classes, objects, loops, and conditional statements. It is user-friendly, saves time, and reduces calculation errors. This application can be used as a console-based program or can be extended to a graphical user interface for better usability. Overall, it provides an easy and efficient solution for income tax estimation.

The system is developed using Core Java concepts such as classes, objects, loops, and conditional statements. It follows a simple input-process-output structure, making it easy to use and understand. The program can be implemented as a console-based application and can also be extended to a graphical user interface for better user interaction.

1.INTRODUCTION

The Income Tax Estimation and Deduction Calculation System is a software application developed to simplify the calculation of income tax. Calculating tax manually can be difficult and time-consuming due to different tax slabs and deduction rules. This system helps users by allowing them to enter their income and deduction details, and it automatically calculates the taxable income and final tax amount.

This system is designed to overcome these problems by providing an automated solution. It allows users to enter their total income and applicable deductions such as investments, insurance, and savings. The system then calculates the taxable income and applies the correct tax rates to determine the final tax amount.

The application is built using Core Java concepts like classes, objects, loops, and conditional statements. It follows a simple input-process-output structure, making it easy to understand and use. The system reduces manual effort, minimizes errors, and provides quick results.

1.1Objective

The main objective of the Income Tax Estimation and Deduction Calculation System is to calculate income tax easily and accurately based on the user's income and deductions. The system aims to reduce manual effort, save time, and minimize calculation errors. It also helps users understand their taxable income and final tax amount in a simple and clear way.

- **To develop a tax calculation system using Java**
- **To automate tax estimation process**
- **To apply OOP concepts**
- **To improve accuracy and efficiency**

1.2Overview

The Income Tax Estimation and Deduction Calculation System is a Java-based application designed to calculate an individual's income tax easily. The system takes the user's income and deduction details as input, computes the taxable income, and applies the appropriate tax rate to generate the final tax amount. It helps reduce manual errors, saves time, and provides accurate results. This system is useful for individuals to quickly estimate their tax in a simple and efficient way.

- **Java-based application for income tax calculation**
- **Takes income and deduction as input**
- **Calculates taxable income automatically**
- **Applies tax rate based on slabs**

1.3 Features

The Income Tax Estimation and Deduction Calculation System is designed to provide a fast, accurate, and user-friendly way to calculate income tax using Java. It allows users to input their total income and various deductions, automatically calculates the taxable income, and applies the appropriate tax rates based on predefined slabs. The system minimizes manual errors and speeds up the entire calculation process. It also includes basic input validation to handle incorrect entries and ensures reliable results. The application is simple to use, making it suitable for beginners as well as general users. Additionally, it can be easily modified or enhanced by adding new tax rules, a graphical user interface, or database support for better functionality and future scalability.

- **Automatically calculates taxable income**
- **Applies tax rates based on slabs**
- **Provides quick and accurate results**
- **Reduces manual calculation errors**
- **Simple Java implementation using OOP concepts**
- **Easy to modify and extend (GUI, database, etc.)**

2.PROGRAM ANALYSIS

The program is developed using Java to calculate income tax based on user input. It takes income and deduction values from the user and computes the taxable income by subtracting deductions from total income. Using simple if-else conditions, the program applies the appropriate tax rate according to different tax slabs. Finally, it displays the calculated tax amount. The program is easy to understand, reduces manual errors, and provides accurate results quickly.

- The program is developed using Java programming language
- It is designed to calculate income tax based on user input
- The user enters income and deduction values
- The system calculates taxable income (Income – Deduction)
- It applies tax rates using conditional statements (if-else)

2.1 Problem Definition

Manual calculation of income tax is often difficult, time-consuming, and prone to errors due to complex tax rules and deductions. Many individuals find it challenging to accurately determine their taxable income and the correct tax amount. Therefore, there is a need for a simple and efficient system that can automate the process of income tax calculation, reduce errors, and provide quick and accurate results based on user input.

- Manual tax calculation is complex and time-consuming
- Difficult to apply correct tax slabs
- Errors may occur in calculations
- Hard to manage deductions properly
- Lack of simple tools for tax estimation
- Need for an automated system

2.2 Project Overview

- Accepts user income and deduction details
- Calculates taxable income automatically
- Applies tax rates based on predefined slabs
- Displays final tax amount clearly
- Reduces manual errors and saves time
- Easy to use and understand
- Useful for individuals and students

2.3 Scopes and Limitations

- Scope:
- Calculates income tax
- Easy to use
- Gives quick results
- Limitations:
- Uses basic tax rules
- No GUI
- No data saving

The system is designed to calculate income tax based on user input such as total income and deductions, providing quick and accurate results. It is useful for individuals to estimate their tax easily without manual calculations. However, the system has some limitations. It considers only basic tax rules and may not include all real-time government updates or complex tax scenarios. It does not support advanced features like online filing, database storage, or integration with official tax systems. The accuracy of the result depends on the correctness of the user input, and the system may need updates when tax laws change.

Scope Highlight:

- Automates income tax and deduction calculation
- Supports different income levels and tax slabs
- Ensures accuracy and reduces manual errors
- Provides quick and clear tax results
- Can be implemented using Core Java (Console/GUI)
- Scalable for future enhancements like database and web integration

3.SYSTEM DESIGN

The system design of the Income Tax Estimation and Deduction Calculation System follows a simple input-process-output structure. The input module collects the user's income and deduction details. The processing module calculates the taxable income by subtracting deductions from the total income. Then, the tax calculation module applies the appropriate tax rate based on predefined slabs. Finally, the output module displays the calculated tax amount to the user. This simple design makes the system easy to understand, implement, and use.

3.1 Modules

The system has different modules to perform each task. The input module gets income and deduction from the user. The processing module calculates taxable income. The tax module calculates the tax amount. Finally, the output module shows the result.

1. **InputModule**

   - Accepts income and deduction details from the user

2. **GradeConversionModule**

   - Calculates taxable income (Income – Deduction)

3. **Performance Module**

   - Applies tax rate based on slabs

4. **OutputModule**

   - Displays the calculated tax

5. **Error Handling Module**

   - Handles invalid inputs and shows error messages

4.METHODOLOGY

The methodology describes the step-by-step process followed to design and develop the Income Tax Estimation and Deduction Calculation System.

- Requirement Analysis

  - Identify the need for calculating income tax and deductions.
  - Understand required inputs such as income, deductions, and tax rates.

---

2. **System Design**

  - Divide the system into modules:

    - Input Module
    - Validation Module
    - Tax Calculation Module
    - Deduction Calculation Module
    - Output Module

  - Plan the workflow of the system.

---

3. **Data Collection**

  - Collect input data from the user:

    - Total income

- Display calculated tax clearly.
- Show final result to the user.

---

4. **Processing**

- Convert user input into usable values.
- Apply formulas to calculate:
  - Taxable Income (Income – Deductions)
  - Applicable Tax based on slabs

---

5. **Implementation**

- Develop the program using Java.
- Use functions/methods for each module.
- Apply logic for calculations and conditions.

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6. **Testing**

- Test with different inputs:
  - Valid inputs
  - Invalid inputs
- Ensure correct GPA and CGPA results.

---

7. **Result & Output**

- The system successfully calculates income tax based on user input.
- Shows appropriate messages for incorrect inputs.

4.1. Problem Analysis

Income tax calculation is often done manually, which makes the process complex and time-consuming for many individuals. People may find it difficult to correctly calculate taxable income after applying various deductions, and there is a high chance of errors when applying tax rates and rules. Additionally, tax regulations can be confusing, especially for users without proper knowledge of financial calculations. This creates a need for a simple and automated system that can accurately compute income tax based on user inputs, reduce mistakes, and provide quick and clear results.

4.2. System Design

The system design explains the structure and working of the Income Tax Estimation and Deduction Calculatio System. It is designed to take user inputs such as income and deductions, process them using predefined formulas and conditions, and generate accurate tax results. The system follows a simple and logical flow to ensure ease of use and efficiency.

Modules in System Design

- User Input Module
- Validation Module
- Tax Calculation Module
- Output Module

- Deduction Module
- User Interface Module

System Workflow

- Start the system
- Get total income from user
- Get deduction details from user
- Check if inputs are valid
- If invalid, show error message
- Calculate taxable income (Income – Deductions)
- If taxable income < 0, set it to 0
- Apply tax rates based on income slabs
- Calculate total tax amount

Design Approach

- Use efficient algorithms for quick calculation
- Maintain accuracy and reliability in results
- Design for easy maintenance and modification

.4.3. Program Development

Program development is the process of creating the Income Tax Estimation System step by step. First, the problem and requirements are identified. Then, the program structure, algorithm, and flowchart are designed. After that, the coding is done using Java, where inputs like income and deductions are taken from the user. The system validates the input and performs tax calculation based on rules. The program is then compiled, tested with different values, and errors are fixed. Finally, the program is improved and documented for easy understanding and future use.

Development Details

- Language Used: **Java**
- Platform: Console-based application
- Approach: Modular programming using methods

Working of the Program

1.The program asks the user to enter:

- Total income
- Deduction amounts (savings, insurance, etc.)

- The system validates the input values.
- The program calculates:
- Taxable income (Income – Deductions)
- Tax amount based on income slabs
- The result is displayed along with tax details.

Features of the Program

- Simple and user-friendly interface
- Accurate and fast calculation
- Supports multiple deductions
- Reduces manual errors

Testing

The program is tested using different types of inputs:

- **Valid Inputs**

  - Correct marks/grades and credits
  - Proper number of subjects

- **Invalid Inputs**

  - Negative marks or credits
  - Marks beyond valid range
  - Incorrect data types (letters instead of numbers)

- **Boundary Testing**

  - Minimum and maximum marks (e.g., 0 and 100)
  - Edge grade values

Validation

- Ensures input values are within acceptable limits
- Verifies correct grade conversion
- Displays error messages for invalid inputs

Result of Testing

- The system produces **accurate and consistent results**
- Errors are handled properly
- The program works efficiently for different input cases

4.5. Result and Documentation

- It provides quick results along with **performance analysis**, making it easy for users to understand their academic progress.
- The program works efficiently for different inputs and handles errors properly.

4.6. Algorithm

Start the program

1. **Input user details**

- **Input user details**
- **Input annual income**
- **Input age category**
- **Input financial year**

2.Initialize variables

- **grossIncome = 0**
- **totalDeductions = 0**
- **taxableIncome = 0**
- **taxAmount = 0**

3.Input income sources

- **Salary income**
- **Business/other income**
- **Interest income**
- **Rental income**
- **Add all to grossIncome**

4.Input deductions

- **Section 80C (LIC, PF, etc.)**
- **Section 80D (medical insurance)**
- **Section 80E (education loan)**
- **Other deductions (if any)**
- **Add all to totalDeductions**

5.Calculate taxable income

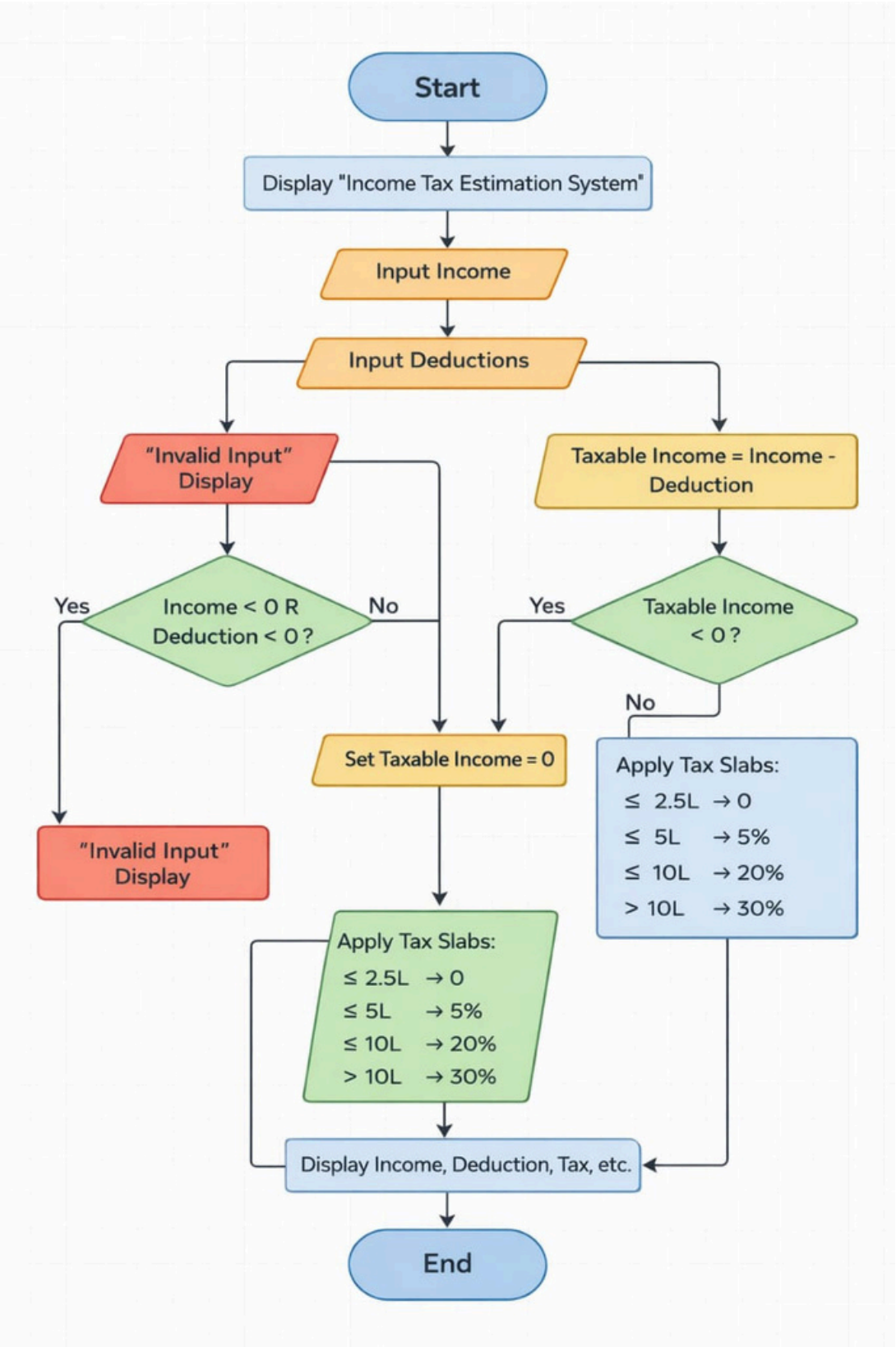
- **Taxable Income = Gross Income − Total Deductions**

6.Apply tax slabs

- **If income ≤ basic exemption → tax = 0**
- **Else apply slab rates step-by-step:**
- **5% for next slab**
- **10% / 20% / 30% (as per rules)**
- **Calculate taxAmount**

7.End the program

4.7. **Flowchart**



5.SOURCE CODE

```
import java.util.Scanner;

public class IncomeTaxSystem {

// Method to calculate tax
public static double calculateTax(double taxableIncome) {
double tax = 0;

if (taxableIncome <= 250000) {
tax = 0;
}
else if (taxableIncome <= 500000) {
tax = (taxableIncome - 250000) * 0.05;
}
else if (taxableIncome <= 1000000) {
tax = 12500 + (taxableIncome - 500000) * 0.20;
}
else {
tax = 112500 + (taxableIncome - 1000000) * 0.30;
}

return tax;
}

public static void main(String[] args) {
Scanner sc = new Scanner(System.in);

double income = 0;
double deduction = 0;
double taxableIncome = 0;
double tax = 0;

}        System.out.println("===== Income Tax Estimation System =====");
}
// Input Income
System.out.print("Enter Total Income: ");
income = sc.nextDouble();

// Input Deductions
System.out.print("Enter Total Deductions: ");
deduction = sc.nextDouble();

// Validation
if (income < 0 || deduction < 0) {
System.out.println("Invalid input! Values cannot be negative.");
}
```

```
else {

:

// Process

taxableIncome = income deduction;

if (taxableIncome < 0) {

}

taxableIncome = 0;

// Tax Calculation

tax = calculate Tax(taxableIncome);

// Output

System.out.println("\n--Result --");

System.out.println("Total Income: + income);

System.out.println("Total Deduction: " + deduction);

System.out.println("Taxable Income: "+taxableIncome);

System.out.println("Tax Payable :" +

tax);

}

sc.close();

}

}
```

6.INPUT AND OUTPUT SCREENS

```
===== Income Tax Estimation System =====
Enter Total Income: 800000
Enter Total Deductions: 100000

----- Result -----
Total Income     : 800000.0
Total Deduction  : 100000.0
Taxable Income   : 700000.0
Tax Payable      : 52500.0
```

Input

Enter Total Income: 400000
Enter Total Deductions: 50000
Calculation
Taxable = 350000
Tax = (350000 − 250000) × 5% = 5000
Output

```
----- Result -----
Total Income     : 400000.0
Total Deduction  : 50000.0
Taxable Income   : 350000.0
Tax Payable      : 5000.0
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The Income Tax Estimation and Deduction Calculation System works effectively by calculating tax based on user input. It simplifies the complex process of manual tax calculation and provides accurate results using predefined tax slabs. The system is easy to use and saves time for users. During testing, it handled different inputs correctly and produced reliable outputs. However, the system depends on fixed tax rules, so it needs updates when tax policies change.

- Accurate tax calculation
- Faster than manual method
- Easy to use
- Challenges:
- Understanding tax slabs
- Handling edge cases

7.2 Conclusion

The Income Tax Estimation and Deduction Calculation System successfully calculates income tax in a simple and accurate way. It reduces manual work, saves time, and minimizes errors. The system is easy to use and helps users understand their tax clearly. Overall, the project achieves its objective and can be improved further by adding more features and updating tax rules.

- Project successfully calculates income tax
- Reduces errors and saves time
- Demonstrates Java and OOP concepts

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